

ME 333
MECHANICS OF MATERIALS

Lecture 1: Introduction
Topic : Course Outline & Course Policy

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Sem I, 2025 - 2026

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Course Proceedings

- Lectures: Wed & Fri (10:00 am - 11:20 am) - AB 7/109
- Lab Sessions: Mon (2:00 pm - 3:20 pm) - AB 4/101
First lab on 11th August (alternates every other week with tutorials).
- Tutorials: Mon (2:00 pm - 3:20 pm) - AB 7/109 (First tutorial on 18th August)

Mode of Communication :

- Google Classroom - Class code: lluyxsvu

Instructor & Tutor :

- Prof. Ravi Sastri Ayyagari (ravi.ayyagari@iitgn.ac.in; AB 4/309)

Lab Instructor :

- Prof. Harmeet Singh (harmeet.singh@iitgn.ac.in; AB 4/301)

Principal Assistant : Dr. Kuldeep Yadav

TAs : Parasuramuni Vishnu, Basharat Shafi, Nikhil Kaser, Aaditya Paresh Nimkar, Abhinab Sharma, Md Riaz Ali Mamud, Sivam Behera, Yash Kashiv

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- 2 Exams (Mid-term Sep 19* & Final Nov 25*).
- 8 – 10 Practice Problem sets (submission not required).
- 25% Lab Component, 25% Quizzes (4), 25% Mid-term exam, 25% Final exam.
- It is strongly encouraged that students attend all lectures, labs and tutorials.
- Grading Policy: Score of 40% or less is an F grade.
- Usage of mobiles, head/ear phones and laptops is not allowed during lectures and tutorials. Bring your own calculators for tutorials and exams.
- Don't be late to lectures, labs or tutorials (< 5 mins permitted).
- Plagiarism in any form guarantees an F. Please read the Student Honor Code[†].

* Tentative Dates

[†] <https://iitgn.ac.in/students/honourcode>

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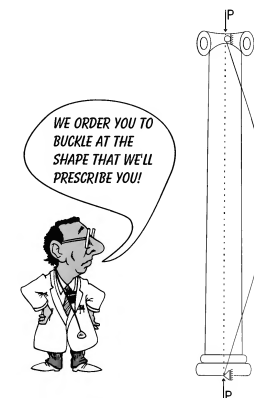
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Q Why another course after Mechanics of Solids ?



Ref: Elishakoff I., CRC Press, 2004

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What is fundamentally different between ES221 and ME333 ?

- ⑥ Formulation of elasticity problem, boundary conditions, plane stress and plane strain problems. Airy stress function, solution by polynomials - [\[Practice Problems 9\]](#)
- ⑦ 2-D problems in polar cylindrical coordinates, equilibrium equations and strain-displacement relations, thick cylinders, composite tubes, stress distribution around a circular hole, stress concentration factor - [\[Practice Problems 10\]](#)

Text : *Advanced Mechanics of Materials* by Arthur P. Boresi, Richard J. Schmidt, Sixth Edition. [Wiley India Pvt. Ltd.](#)

Course Outline

- ① Strain at a point, strain-displacement relations in Cartesian coordinates, compatibility equations - [\[Practice Problems 1\]](#)
- ② Stress at a point, equilibrium equations, 3-D state of stress, principal stresses and maximum shear stress, stress invariants - [\[Practice Problems 2\]](#)
- ③ Stress-Strain relations, Concept of yielding - [\[Practice Problems 3 & 4\]](#)
- ④ Torsion & Torsion of non-circular shafts - [\[Practice Problems 5\]](#)
- ⑤ Bending, unsymmetrical bending, shear flow and shear center; Symmetric bending of curved beams - [\[Practice Problems 6, 7 & 8\]](#)

References

- ① S. P. Timoshenko, *Strength of Materials Part 2: Advanced Theory and Problems*; Third Edition, CBS, 2002.
- ② J. R. Barber, *Intermediate Mechanics of Materials*; Second Edition, Springer, 2010.
- ③ Ansel C. Ugural, Saul K. Fenster, *Advanced Mechanics of Materials and Applied Elasticity*; Fifth Edition, Prentice Hall, 2011.
- ④ R.D. Cook and W.C. Young, *Advanced Mechanics of Materials*; Second Edition, Prentice Hall, 1999.
- ⑤ E. P. Popov, *Engineering Mechanics of Solids*; Second Edition, Pearson, 1998.

Reading Assignment

Review concept of Strain